

Title: Git Hands-On Lab 1 - Basic Git Commands and Setup**Problem Statement :****Objectives:** Familiar with Git commands like git init, git status, git add, git commit, git push, and git pull.

In this hands-on lab, you will learn how to:

- Setup your machine with Git Configuration
- Integrate notepad++.exe to Git and make it a default editor
- Add a file to source code repository
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Prerequisites:

- Install Git Bash client in your machine

Step 1:

- Setup your machine with Git Configuration
 - To create a new repository, signup with GitLab and register your credentials. Login to GitLab and create a “GitDemo” project.
1. To check if Git client is installed properly: Open Git bash shell and execute git --version.
I have done this step.

```
pchai@Chaitanya MINGW32 ~
$ git version
git version 2.48.1.windows.1
```

2. To configure user level configuration of user ID and email ID execute git config --global user.name "Your Name" and git config --global user.email "your.email@example.com".
I have done this step.

```
pchai@Chaitanya MINGW32 ~
$ git config --global user.name "Chaitanya"

pchai@Chaitanya MINGW32 ~
$ git config --global user.email "p.chaitanyapyla@gmail.com"
```

3. To check if the configuration is properly set, execute the following command: git config --list.
I have done this step.

```
pchai@Chaitanya MINGW32 ~
$ git config --global --list
user.name=Chaitanya
user.email=p.chaitanyapyla@gmail.com
core.editor=notepad++.exe -multiInst -nosession
```

Step 2: Integrate notepad++.exe to Git and make it a default editor

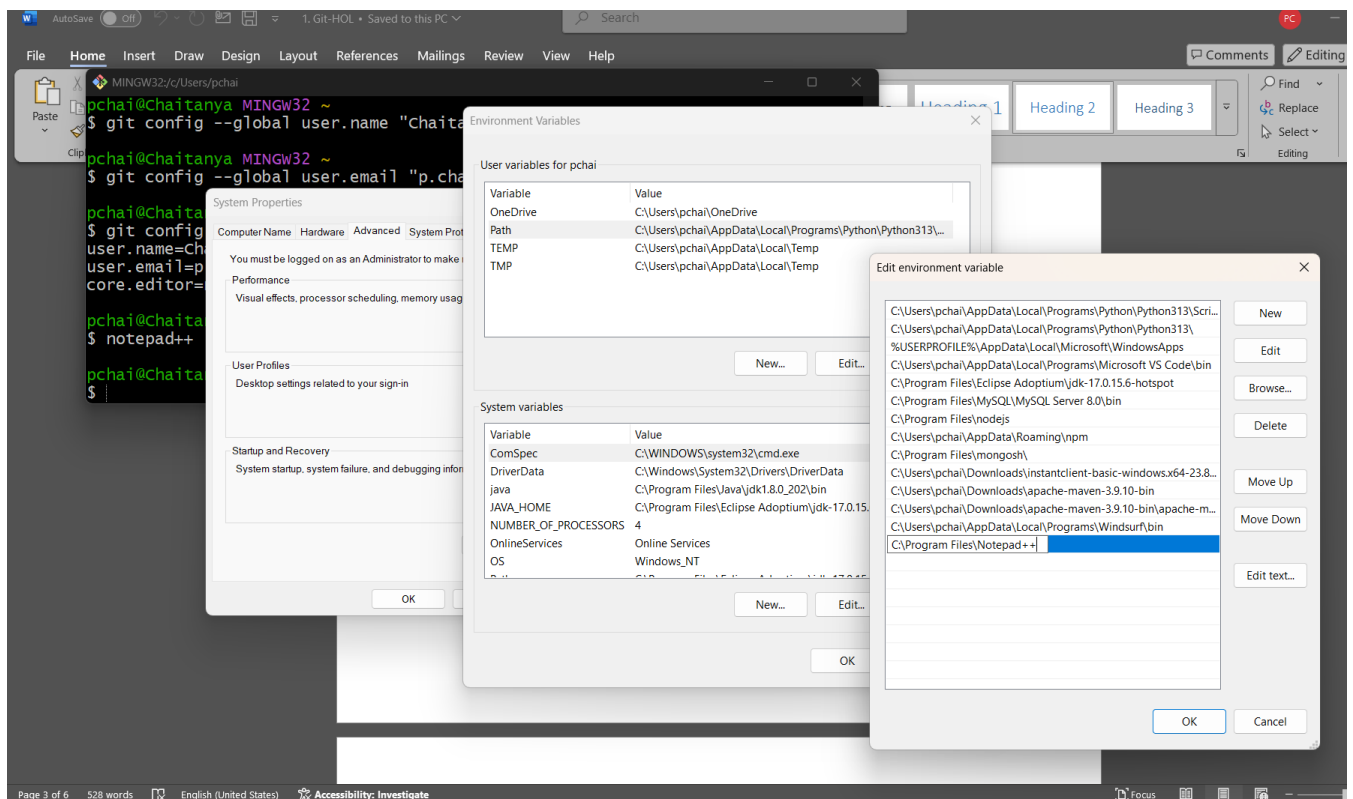
1. To check if notepad++.exe executes from Git bash: notepad++.

```

MINGW32/c/Users/pchai
pchai@Chaitanya MINGW32 ~
$ notepad++
bash: notepad++: command not found

```

If Git bash could not recognize notepad++ command, add path of notepad++.exe to the environment path variable.



2. Exit Git bash shell, open bash shell and execute notepad++. Now, notepad++ will open from Git bash shell.

```

pchai@Chaitanya MINGW32 ~
$ notepad++
pchai@Chaitanya MINGW32 ~
$ notepad++
pchai@Chaitanya MINGW32 ~
$ notepad++.exe bash -profile

```

3. To create an alias command for notepad++.exe, execute alias np=notepad++. It will open notepad++ from bash shell, and create a user profile by adding the line in notepad++.

```

pchai@Chaitanya MINGW32 ~
$ notepad++.exe bash -profile

```

- To configure the editor, execute the command `git config --global core.editor "notepad++ -multiInst -nosession"`.

```
pchai@Chaitanya MINGW32 ~
$ git config --global core.editor "notepad++.exe -multiInst -nosession"

pchai@Chaitanya MINGW32 ~
$ git config --global -e

pchai@Chaitanya MINGW32 ~
$ git init GitDemo
Initialized empty Git repository in C:/Users/pchai/GitDemo/.git/

pchai@Chaitanya MINGW32 ~
$ ls -al
total 102419
drwxr-xr-x 1 pachai 197609      0 Aug  9 21:23 ./
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 ../
-rw-r--r-- 1 pachai 197609    145 Mar 31 2024 .appletviewer
-rw-r--r-- 1 pachai 197609    160 Aug  9 21:18 .bash_history
drwxr-xr-x 1 pachai 197609      0 Apr  8 2024 .cache/
drwxr-xr-x 1 pachai 197609      0 Jul 24 10:57 .codeium/
drwxr-xr-x 1 pachai 197609      0 Oct 16 2024 .dbclient/
drwxr-xr-x 1 pachai 197609      0 Jul 28 21:21 .docker/
-rw-r--r-- 1 pachai 197609    113 Aug  9 21:21 .gitconfig
drwxr-xr-x 1 pachai 197609      0 Jul  6 15:28 .gradle/
drwxr-xr-x 1 pachai 197609      0 Jan 26 2024 .idlers/
```

Step 3: Add a file to source code repository

- Open Git bash shell and create a new project “GitDemo” by executing the command `mkdir GitDemo && cd GitDemo && git init`.

```
pchai@Chaitanya MINGW32 ~
$ git init GitDemo
Initialized empty Git repository in C:/Users/pchai/GitDemo/.git/
```

- Git bash initializes the “GitDemo” repository. To verify, execute the command `ls -la`. It will display all the hidden files in the Git “working directory”.

```
MINGW32:/c:/Users/pchai
pchai@Chaitanya MINGW32 ~
$ git init GitDemo
Initialized empty Git repository in C:/Users/pchai/GitDemo/.git/

pchai@Chaitanya MINGW32 ~
$ ls -al
total 102419
drwxr-xr-x 1 pachai 197609      0 Aug  9 21:23 ./
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 ../
-rw-r--r-- 1 pachai 197609    145 Mar 31 2024 .appletviewer
-rw-r--r-- 1 pachai 197609    160 Aug  9 21:18 .bash_history
drwxr-xr-x 1 pachai 197609      0 Apr  8 2024 .cache/
drwxr-xr-x 1 pachai 197609      0 Jul 24 10:57 .codeium/
drwxr-xr-x 1 pachai 197609      0 Oct 16 2024 .dbclient/
drwxr-xr-x 1 pachai 197609      0 Jul 28 21:21 .docker/
-rw-r--r-- 1 pachai 197609    113 Aug  9 21:21 .gitconfig
drwxr-xr-x 1 pachai 197609      0 Jul  6 15:28 .gradle/
drwxr-xr-x 1 pachai 197609      0 Jan 26 2024 .idlers/
drwxr-xr-x 1 pachai 197609      0 Oct 16 2024 .lemnix/
-rw-r--r-- 1 pachai 197609    20 Aug  9 20:40 .lessht
drwxr-xr-x 1 pachai 197609      0 Jul 25 13:39 .m2/
-rw-r--r-- 1 pachai 197609    129 Mar  2 19:22 .node_repl_history
drwxr-xr-x 1 pachai 197609      0 May 19 2024 .redhat/
drwxr-xr-x 1 pachai 197609      0 Mar 21 07:25 .rest-client/
drwxr-xr-x 1 pachai 197609      0 Feb 26 16:26 .sonarlint/
drwxr-xr-x 1 pachai 197609      0 Oct 20 2023 .vscode/
drwxr-xr-x 1 pachai 197609      0 Jul 24 10:57 .windurf/
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 Appdata/
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 'Application Data' -> /c/Users/p
chai/AppData/Roaming/
drwxr-xr-x 1 pachai 197609      0 Mar 18 22:11 Chaitu_Cricket/
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 Contacts/
drwxr-xr-x 1 pachai 197609    58 Dec 16 2024 Cookies -> /c/Users/pchai/Appda
ta/Local/Microsoft/Windows/InetCookies/
drwxr-xr-x 1 pachai 197609      0 Jul 27 16:52 Desktop/
drwxr-xr-x 1 pachai 197609      0 Jun 25 2023 Documents/
drwxr-xr-x 1 pachai 197609      0 Aug  9 20:46 Downloads/
drwxr-xr-x 1 pachai 197609      0 Mar 20 21:24 Favorites/
drwxr-xr-x 1 pachai 197609      0 Aug  9 21:23 GitDemo/
drwxr-xr-x 1 pachai 197609      0 Aug  9 20:34 IntelGraphicsProfiles/
drwxr-xr-x 1 pachai 197609      0 Dec 16 2024 Links/
```

- To create a file “welcome.txt” and add content to the file, execute the command `echo "Welcome to Git" > welcome.txt`.

```
pchai@Chaitanya MINGW32 ~
$ echo "welcome to the session control">> welcome.txt
```

4. To verify if the file “welcome.txt” is created, execute ls.

```
drwxr-xr-x 1 pchai 197609      0 Mar 18 22:17 react-js-project/
drwxr-xr-x 1 pchai 197609      0 Mar 18 20:52 train-booking/
-rw-r--r-- 1 pchai 197609    31 Aug  9 21:24 welcome.txt
```

5. To verify the content, execute the command cat welcome.txt.

```
pchai@Chaitanya MINGW32 ~
$ cat welcome.txt
Welcome to the session control
pchai@Chaitanya MINGW32 ~
$
```

6. Check the status by executing git status. Now the file “welcome.txt” is available in Git “working directory”.

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git status
On branch master

No commits yet

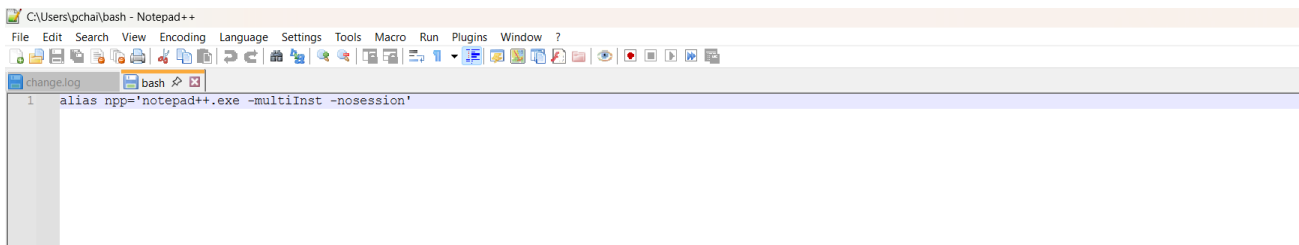
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$
```

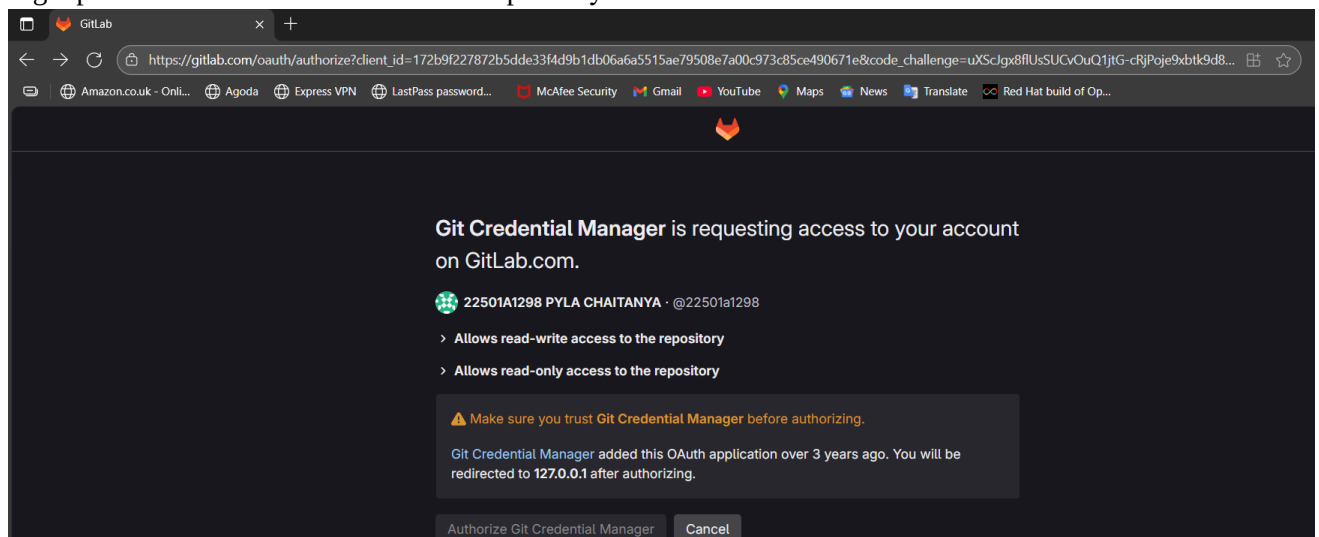
7. To make the file to be tracked by Git repository, execute the command git add welcome.txt.

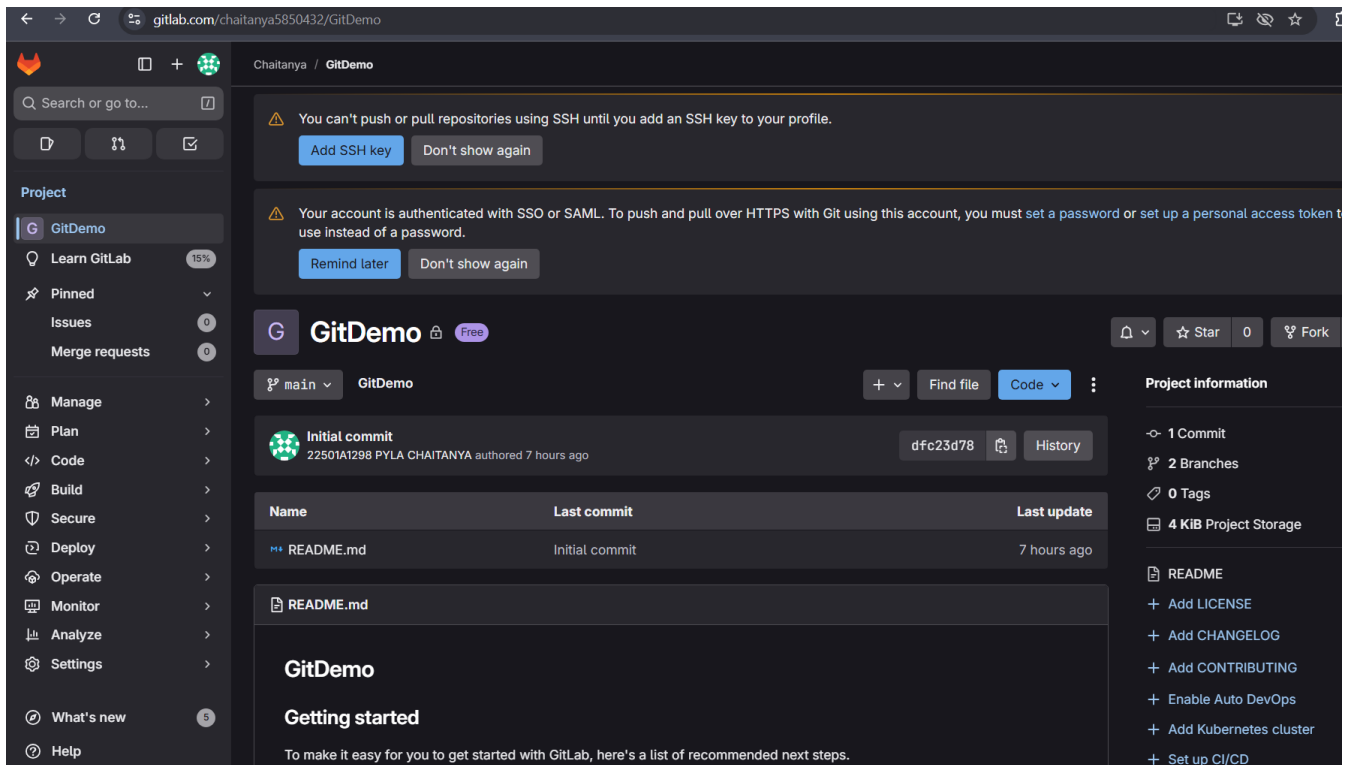
```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF the next time Git touches it
```

8. To add multi line comments, we are opening default editor to comment. Execute the command git commit. Notepad++ editor will open and to add multi-line comment with default editor.



9. Signup with GitLab and create a remote repository “GitDemo”





10. To pull the remote repository, execute `git pull origin master`.

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git commit -m "Add welcome.txt"
[master (root-commit) ff1232a] Add welcome.txt
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ ^C

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git remote add origin https://gitlab.com/chaitanya5850432/GitDemo.git

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git push -u origin master
```

11. To push the local to remote repository, execute `git push origin master`.

